



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

OBJECT-ORIENTED DESIGN IN SMART HEALTHCARE SYSTEMS

CSA1101 – Object Oriented Analysis and Design



Guided By

Dr. A. Gayathri

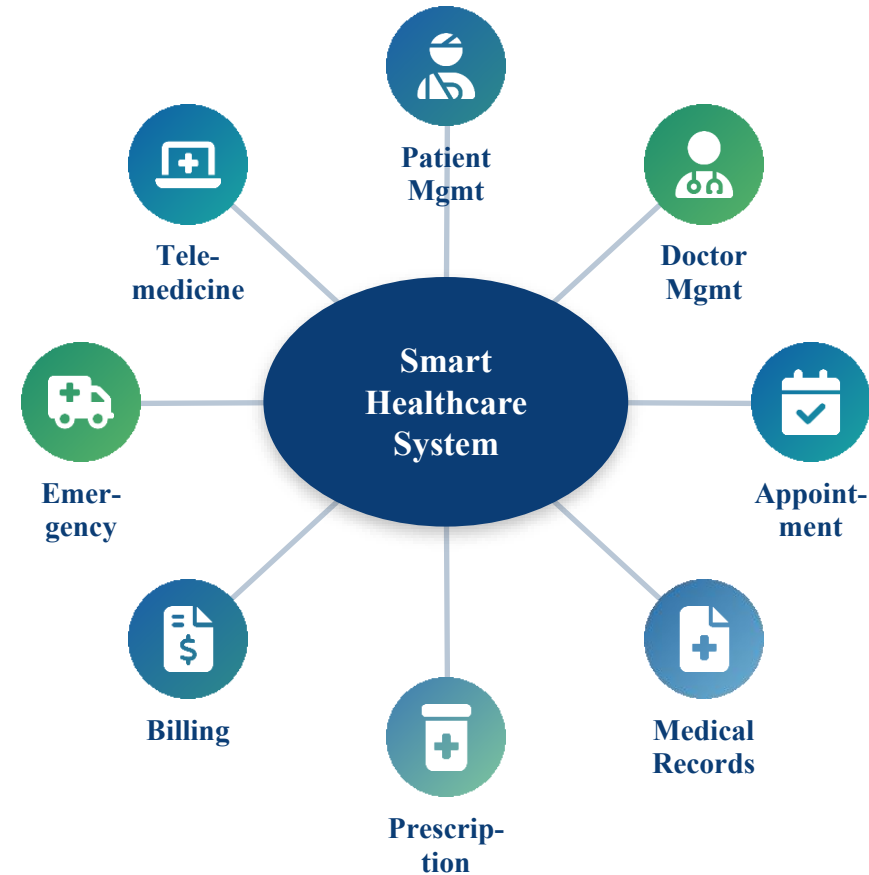
Prepared By

Mathusri P

Register No: 192311363

Introduction to Smart Healthcare System

- A digital ecosystem connecting patients, doctors & hospitals
- Uses OOD, IoT, AI & cloud computing together
- Importance: faster diagnosis, fewer errors, wider access
- Objective: centralize & secure patient data
- Objective: enable real-time doctor–patient interaction
- Objective: automate appointment & billing workflows
- Objective: support remote telemedicine consultations
- Built with modular, reusable Object-Oriented classes



Application of Object-Oriented Concepts

- Class: blueprint defining attributes & methods
- Example – Patient class: patientID, name, diagnosis
- Methods: bookAppointment(), updateRecord()
- Object: a real instance created from a class
- Example: patient1 = new Patient("Aisha")
- Encapsulation: hides internal data using private fields
- Access allowed only via public getters & setters
- Protects sensitive medical data from misuse



Object-Oriented Design

The overall design approach



Class: Patient

Blueprint – ID, Name, Diagnosis



Object: patient1

patient1 = new Patient("Aisha")

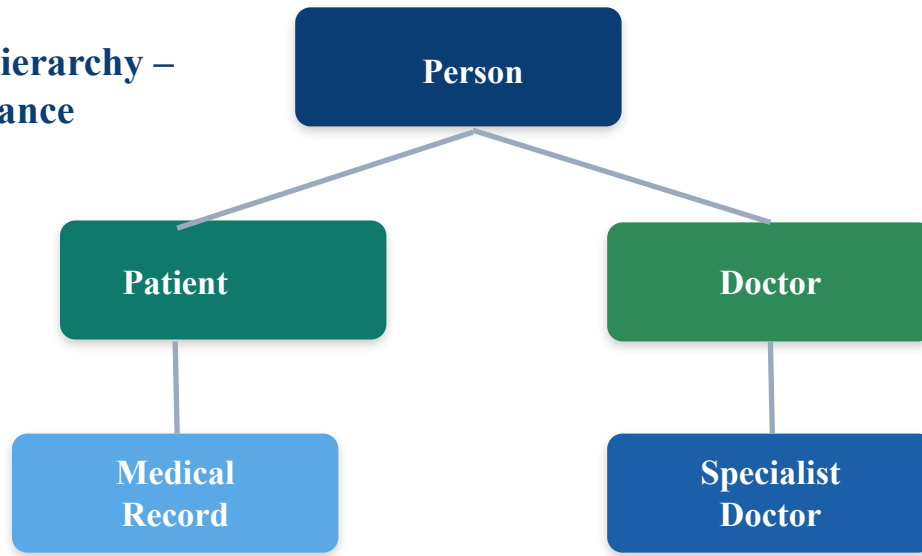


Encapsulation

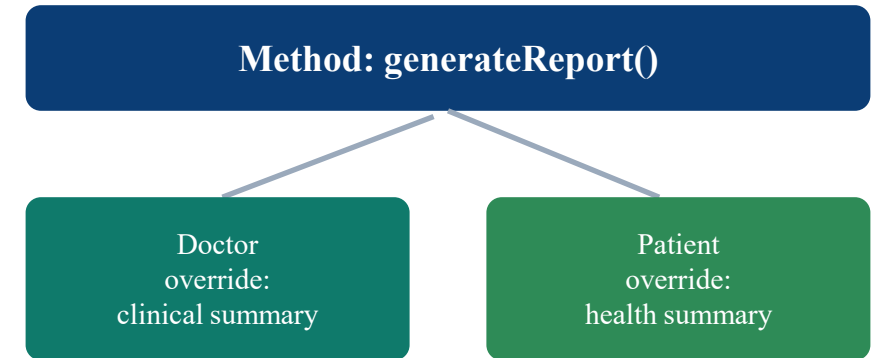
Private data + getters/setters

Inheritance and Polymorphism

UML Class Hierarchy – Inheritance



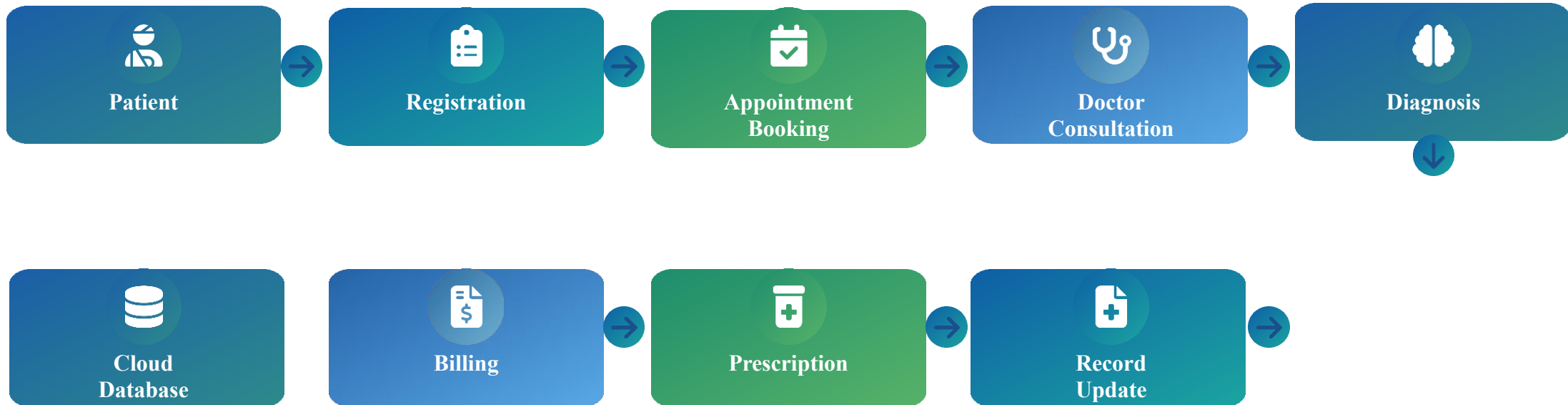
- Inheritance: child class reuses parent class features
- Person is the base class (name, age, ID)
- Patient & Doctor inherit common Person properties
- Specialist Doctor further inherits from Doctor



Polymorphism – generateReport()

- Polymorphism: same method, different behaviour
- Method overriding customizes inherited behaviour
- Dynamic binding resolves the method at runtime
- Example: generateReport() differs for Doctor & Patient

Working Architecture – Patient Journey



- Registration: patient enters personal & medical details
- Appointment Booking: system schedules a doctor visit slot
- Doctor Consultation: physician examines patient & history
- Diagnosis: condition identified using OOD-based logic
- Record Update: EHR updated with the latest visit data
- Prescription: system auto-generates a digital prescription
- Billing: invoice & payment processed automatically
- Cloud Database: all records stored & synced securely

Benefits and Conclusion



Scalability

Handles growing patients
& hospitals easily



Security

Encapsulation & access
control protect data



Maintainability

Modular classes simplify
updates & fixes

Code Reusability

Flexibility

Performance

Future Expansion

Cloud Integration

AI Integration

- Scalability: system grows smoothly with more users & modules
- Security: encapsulation & access control safeguard patient data
- Maintainability: modular classes simplify updates and debugging
- Flexibility: polymorphism adapts to diverse healthcare workflows
- Future-ready: easy integration of AI, IoT & analytics
- Conclusion: OOD principles power robust, scalable smart healthcare systems

THANK YOU